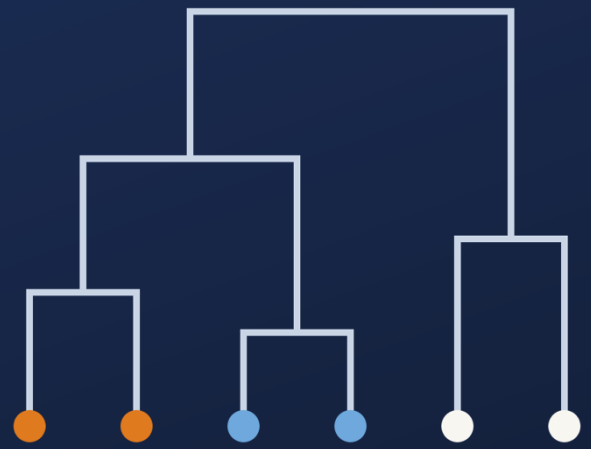
ML & AI FOR LIFE SCIENCES 2026 · DAY 1 · LECTURE 2

Unsupervised learning

Finding groups nobody labelled: k-means, hierarchical and graph clustering



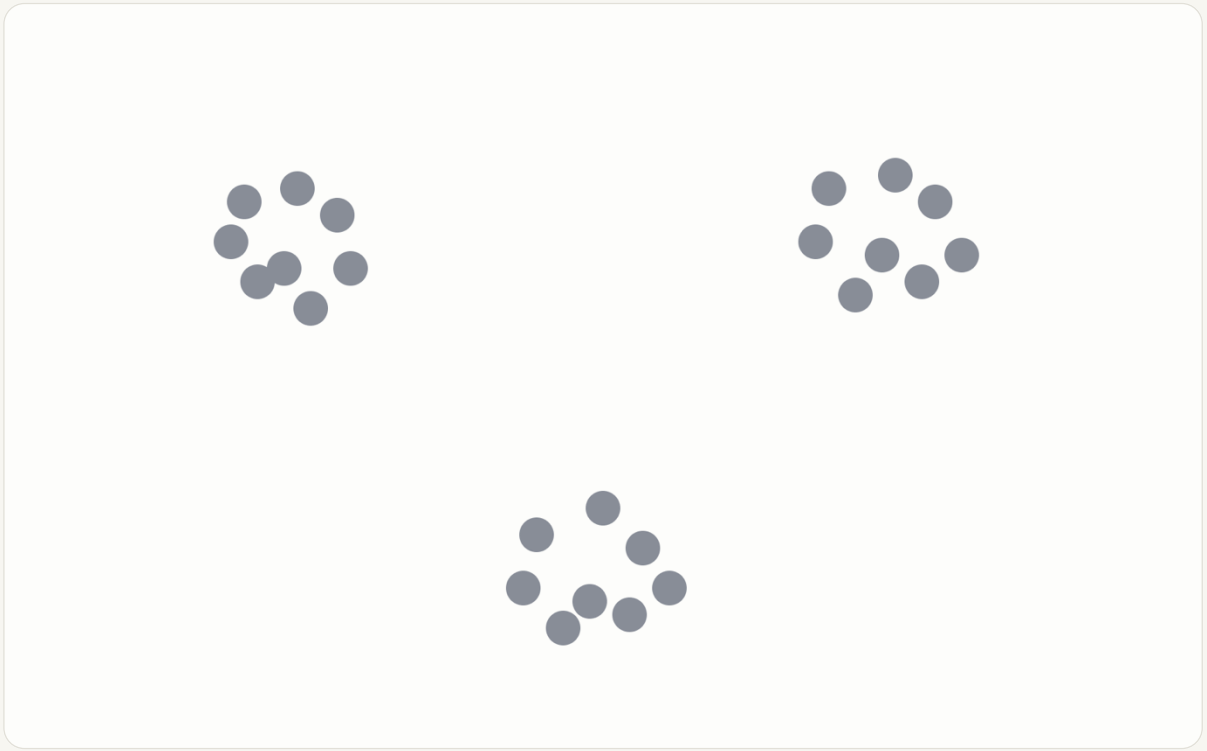
Trần Thượng Vân Du · SIB Swiss Institute of Bioinformatics

VNUHCM-University of Science · 02/11/2026

A PUZZLE TO START

VI: không có nhãn

Here are cells from a tissue. No labels. How many types?



Your eyes found groups in under a second.

In 2 dimensions that is easy. In 20,000 dimensions we need an algorithm.

Clustering = making that intuition precise and repeatable.

DEFINITION

VI: học không giám sát

Unsupervised learning finds structure in X alone

Clustering

Which samples belong together?

Dimensionality reduction

Which few axes capture the variation?

Association & density

Which features co-occur? Which samples are unusual?

No y, so no "right answer" to check against. That makes validation the hard part.

Clustering created several famous classifications

Breast cancer subtypes

Hierarchical clustering of expression → luminal, HER2, basal-like (Perou et al. 2000)

Cell types

Graph clustering of single-cell RNA-seq → cell atlases

Chromatin states

Segmenting the genome by histone marks (ChromHMM, Segway)

Patient subgroups

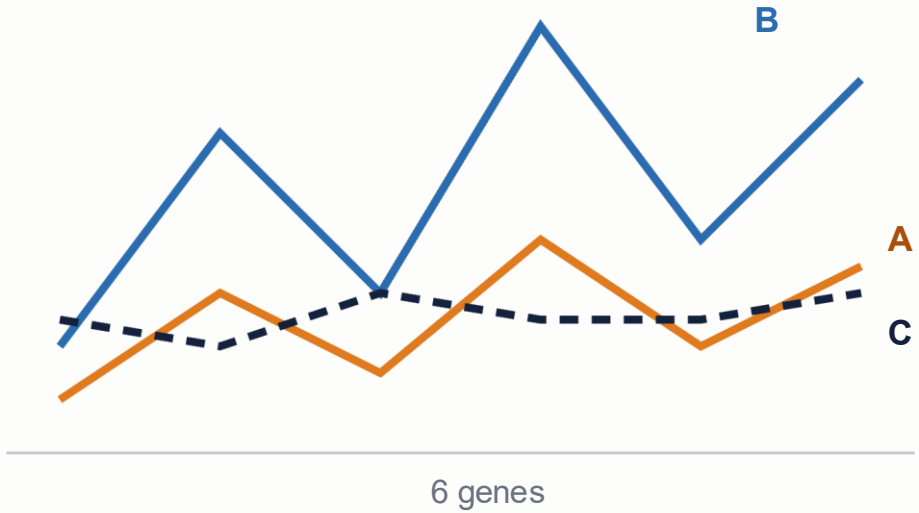
Data-driven type 2 diabetes subgroups from clinical variables (Ahlqvist et al. 2018)

Perou et al. Nature 2000 · Ernst & Kellis Nat Methods 2012 · Ahlqvist et al. Lancet Diabetes Endocrinol 2018

WHAT DOES "SIMILAR" MEAN?

VI: khoảng cách · độ tương đồng

The distance you choose decides the clusters you get



Euclidean distance

Compares levels → A is closest to **C**

Correlation distance (1 − r)

Compares shape → A is closest to **B**

Co-regulated genes share shape, not level: correlation is often the better choice for them.

Without scaling, the loudest feature wins

The problem

Glucose in mg/dL (70–300) and HbA1c in fraction (0.04–0.12).

Distances are almost entirely glucose. HbA1c is ignored.

The fix

log for skewed counts, then **z-score** each feature:

```
z = (x - mean) / sd
```

Every feature now has mean 0, sd 1.

But think first

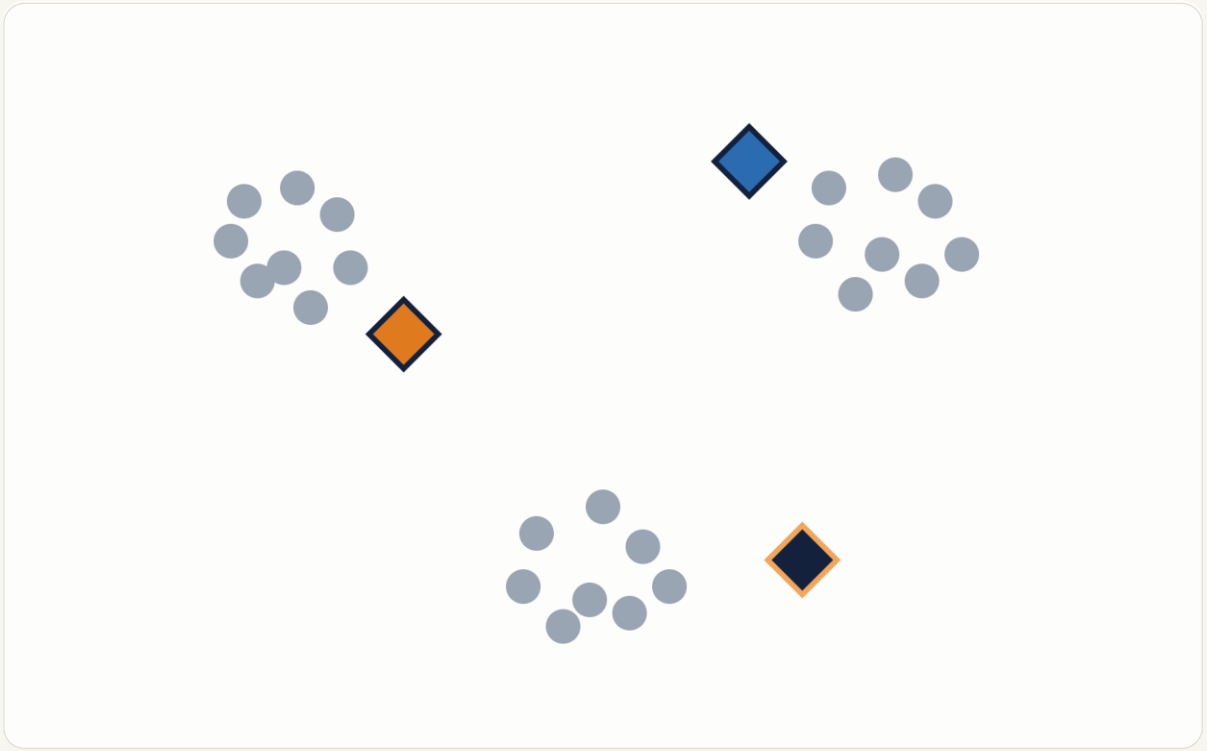
Scaling also inflates noisy, low-expressed genes.

Common practice: keep the top 1,000–5,000 most variable genes, then scale.

K-MEANS · STEP 1 OF 4

VI: tâm cụm (centroid)

Choose k = 3 and drop three centroids at random



1 · Place k centroids at random

2 · Assign each point to the nearest centroid

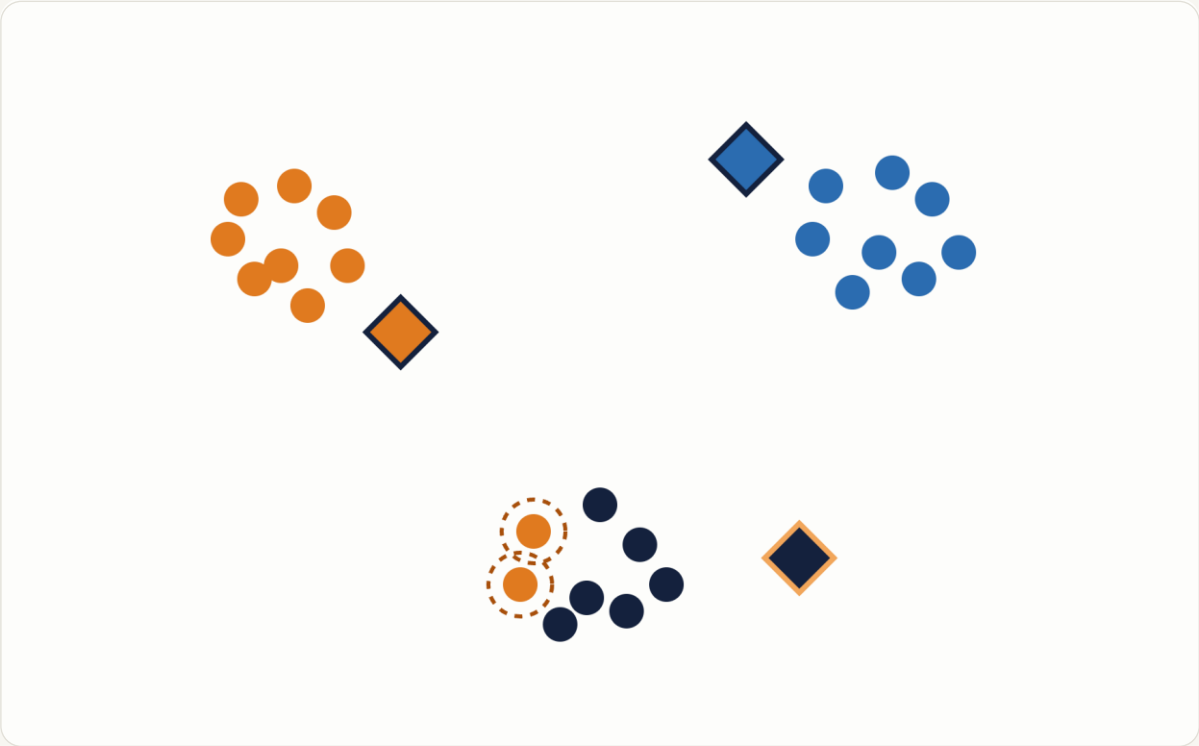
3 · Move each centroid to the mean of its points

4 · Repeat 2–3 until nothing changes

K-MEANS · STEP 2 OF 4

VI: gán điểm vào cụm

Each point joins its nearest centroid



1 · Place k centroids at random

2 · Assign each point to the nearest centroid

3 · Move each centroid to the mean of its points

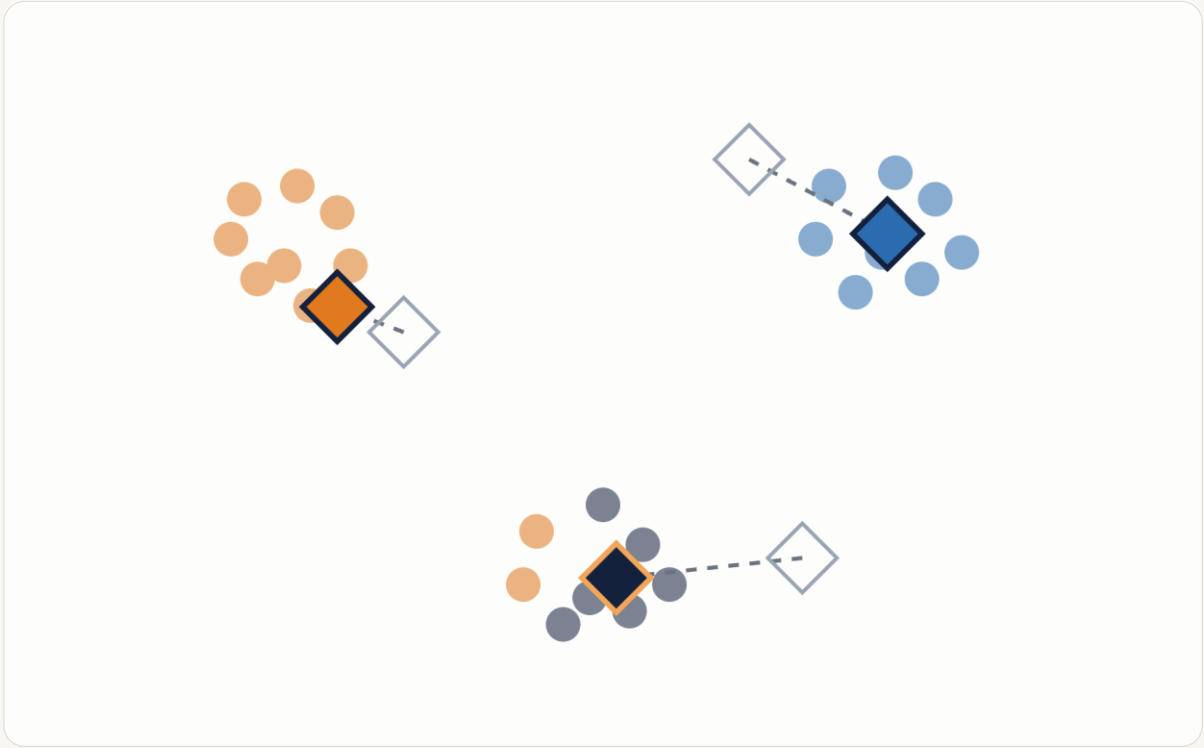
4 · Repeat 2–3 until nothing changes

Circled: two bottom points go to orange, for now.

K-MEANS · STEP 3 OF 4

VI: cập nhật tâm cụm

Each centroid moves to the mean of its points



1 · Place k centroids at random

2 · Assign each point to the nearest centroid

3 · Move each centroid to the mean of its points

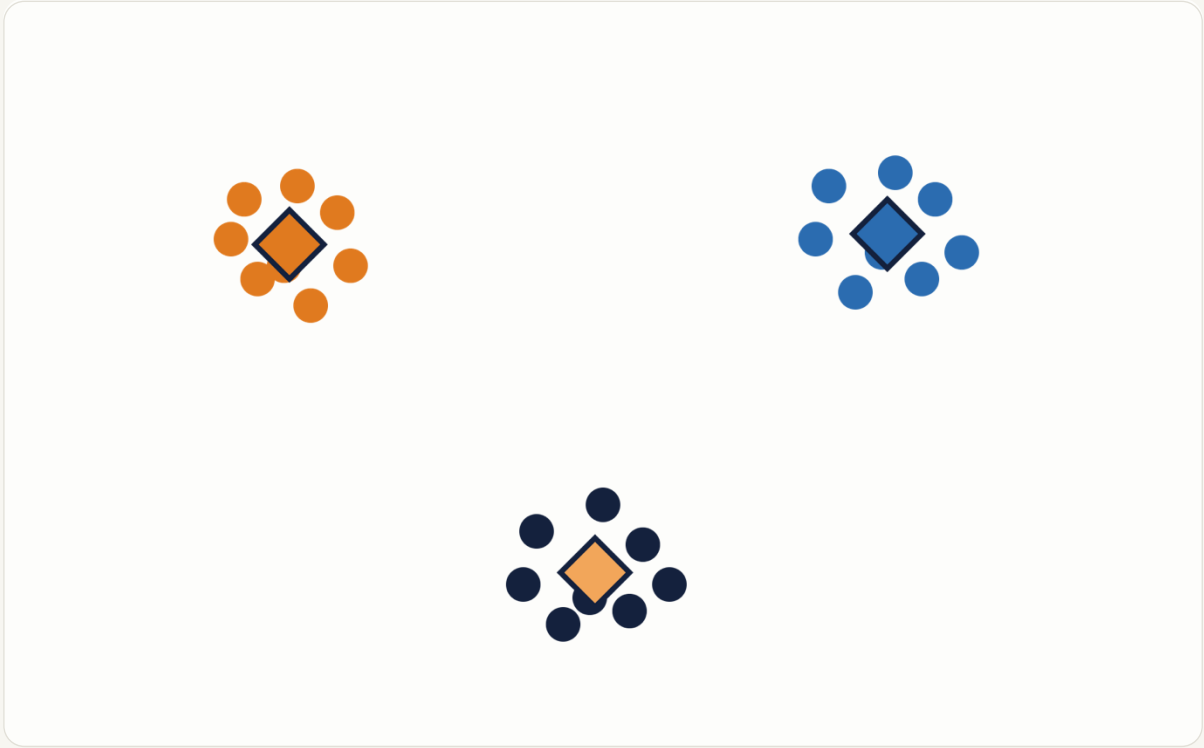
4 · Repeat 2–3 until nothing changes

Outlines = old positions; dashed = movement.

K-MEANS · STEP 4 OF 4

VI: hội tụ

Repeat until no point changes cluster



1 · Place k centroids at random

2 · Assign each point to the nearest centroid

3 · Move each centroid to the mean of its points

4 · Repeat 2–3 until nothing changes

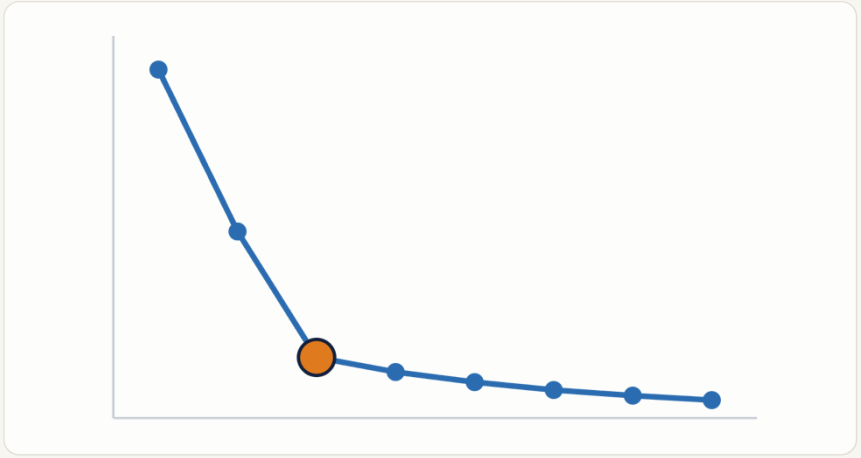
Goal: minimise the total squared distance of points to their centroid (within-cluster sum of squares).

CHOOSING K

VI: chọn số cụm k

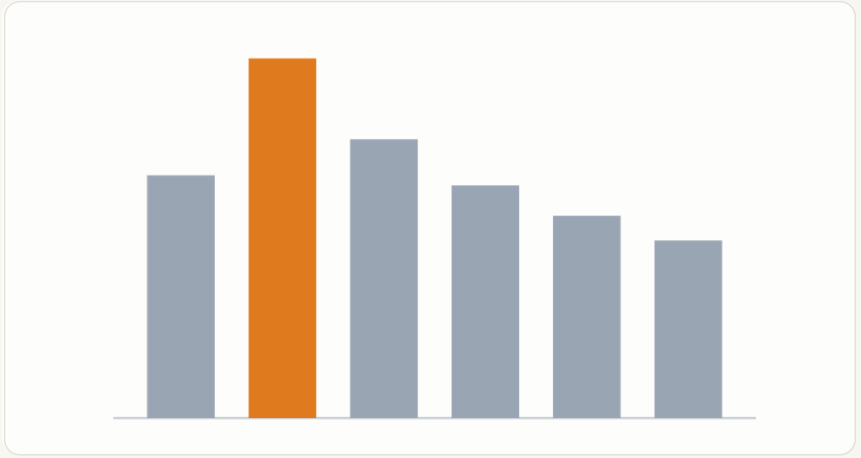
Two quick tools to choose k, plus your biology

Elbow plot



Within-cluster spread vs k (1–8). Pick the bend: **k = 3**.

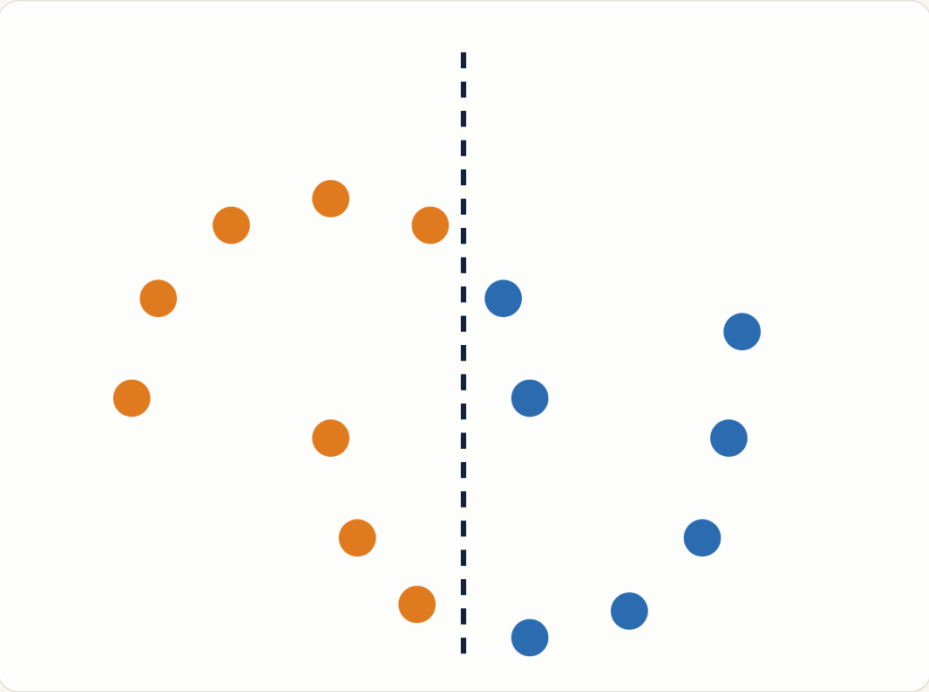
Silhouette score



How well each point fits its own cluster vs the next one (−1 to 1). Highest at **k = 3**.

Real data rarely gives a clean answer. Biology and stability decide too.

k-means assumes round, similar-sized blobs



Orange / blue = k-means split; dashed = its boundary

Shape: two crescents are cut in half.

Random start: different runs can disagree.

Outliers: a single extreme sample pulls a mean.

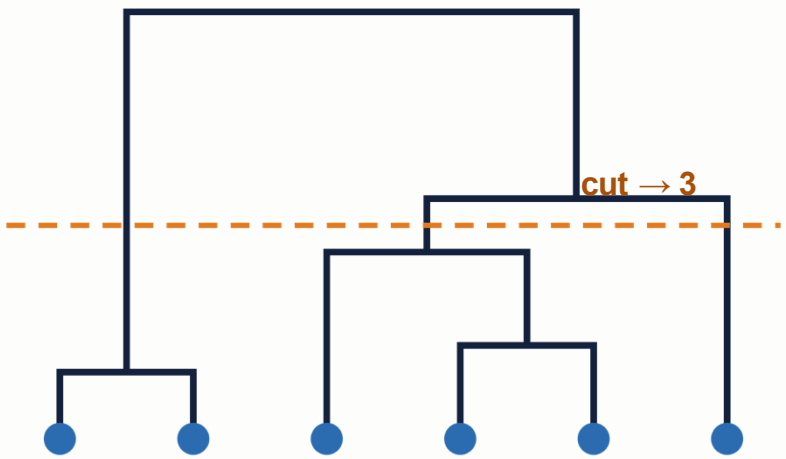
You must give k.

Alternatives: hierarchical, DBSCAN (density), Gaussian mixtures, graph-based.

HIERARCHICAL CLUSTERING

VI: phân cụm phân cấp · cây phân nhánh

Merge the two closest groups, again and again



Start: each sample is its own cluster. Height of a join = distance at which groups merged. **No k needed upfront**: cut the tree where you like.

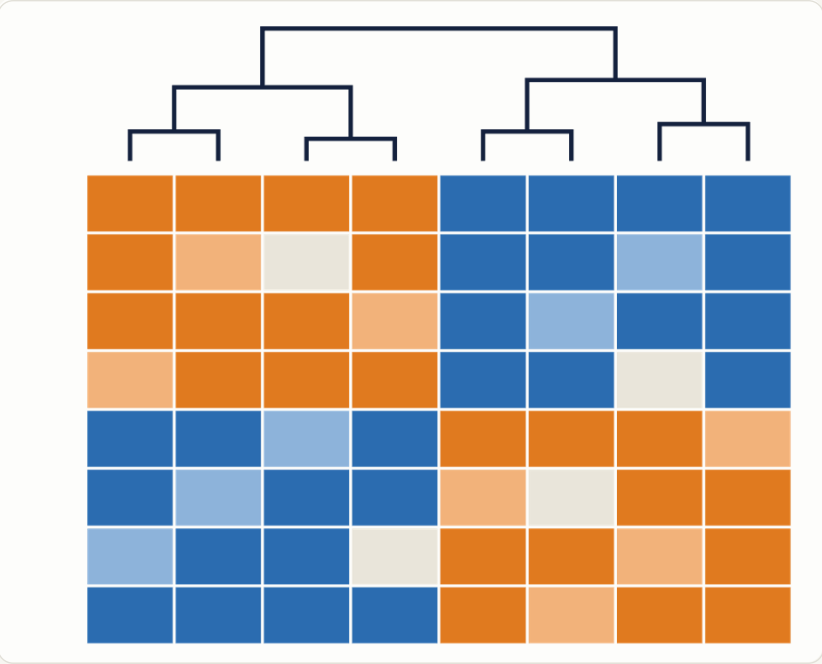
Linkage = distance between groups

Linkage	Uses
Single	Closest pair → long chains
Complete	Farthest pair → compact
Average	Mean of all pairs
Ward	Smallest increase in variance

THE OMICS CLASSIC

VI: bản đồ nhiệt

A clustered heatmap shows samples and genes together



Columns = samples, ordered by the dendrogram on top.

Rows = genes (z-scored); orange = high, blue = low.

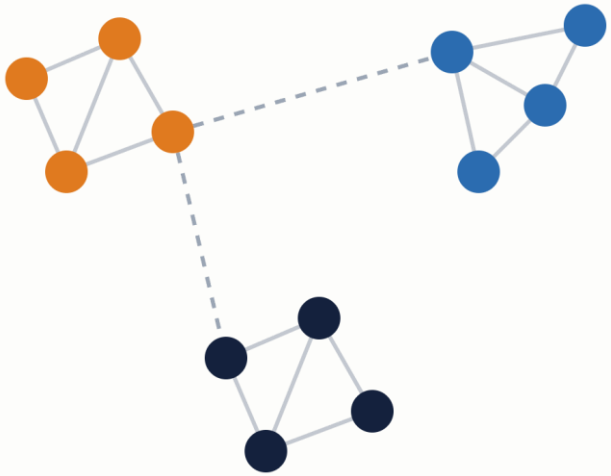
Two sample groups, each with its own gene block: the pattern behind most "subtype" figures.

seaborn.clustermap · R: pheatmap, ComplexHeatmap

GRAPH-BASED CLUSTERING

VI: phân cụm dựa trên đồ thị

Single-cell data is clustered as a network of neighbours



1 · Build the k-nearest-neighbour graph (same as UMAP, Lecture 1).

2 · Find **communities**: groups with many links inside and few between.

Leiden (or Louvain) is the default in Scanpy and Seurat; scales to millions of cells.

Resolution parameter ↑ → more, smaller clusters. Another "k" in disguise.

Traag, Waltman & van Eck, Sci Rep 2019 (Leiden algorithm)

VALIDATION · POLL

VI: các cụm có thật không?

Clustering always returns clusters. Are they real?

Internal

Silhouette: are clusters compact and separated?

Stability

Re-cluster on 80% subsamples: do the same groups come back?

External

Adjusted Rand Index vs known labels, if any (1 = perfect, 0 = chance)

Biological

Known markers? Different outcomes? Replicated in another cohort?

Trust a cluster only when several of these agree.

Beware circular analysis

Double dipping

Cluster cells, then test which genes differ between clusters using the same data.

The clusters were made to differ, so p-values are far too small.

Labels built from features

PAM50 breast cancer subtypes are defined from expression of 50 genes.

Recovering them by clustering expression is partly a circle, not a discovery.

Batch = cluster

If clusters line up with sequencing date or hospital, you clustered the batch.

Always cross-tabulate clusters × metadata.

Zhang, Kamath & Tse, Cell Systems 2019 · Parker et al. J Clin Oncol 2009 (PAM50)

TAKE-AWAYS → COLAB

Three things to remember

1

Distance and scaling choices shape the clusters as much as the algorithm.

2

k-means: fast, needs k, likes round blobs. Hierarchical: a tree you cut.

3

A cluster is a hypothesis: check stability, metadata and biology.